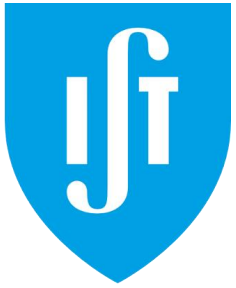
Update of X(3872) Studies in LHC Run 3

Novel probes of the Quark-Gluon Plasma medium

Henrique Legoinha, Lekai yao, Nuno Leonardo, Zhen hu

2026.4.23



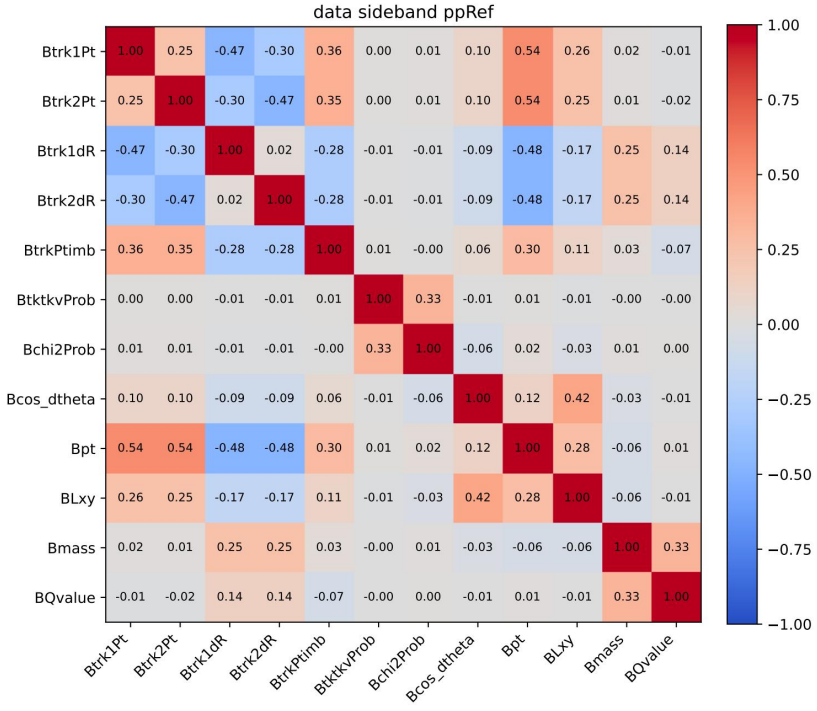


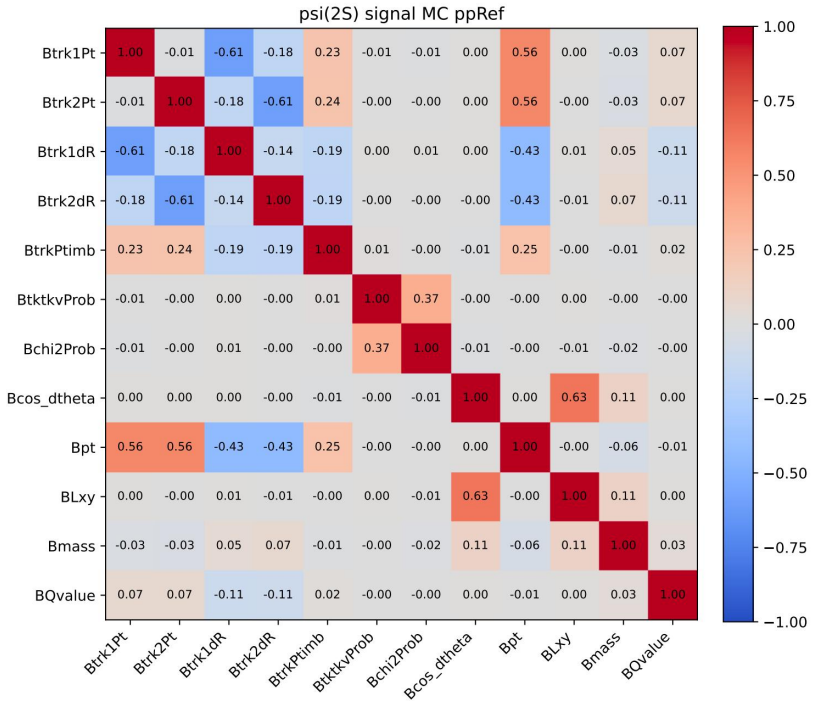


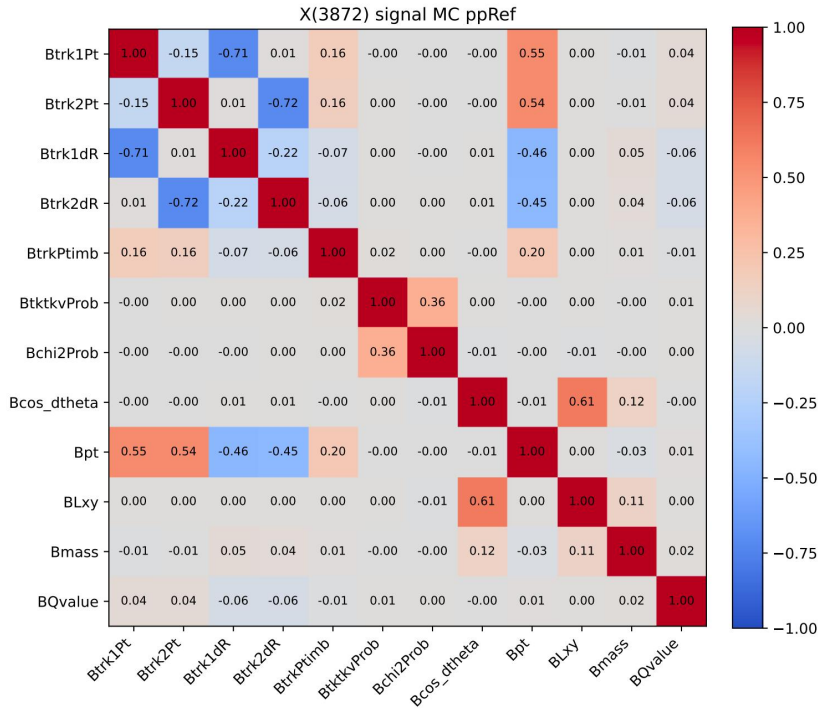




Correlation Matrix for sideband and MC

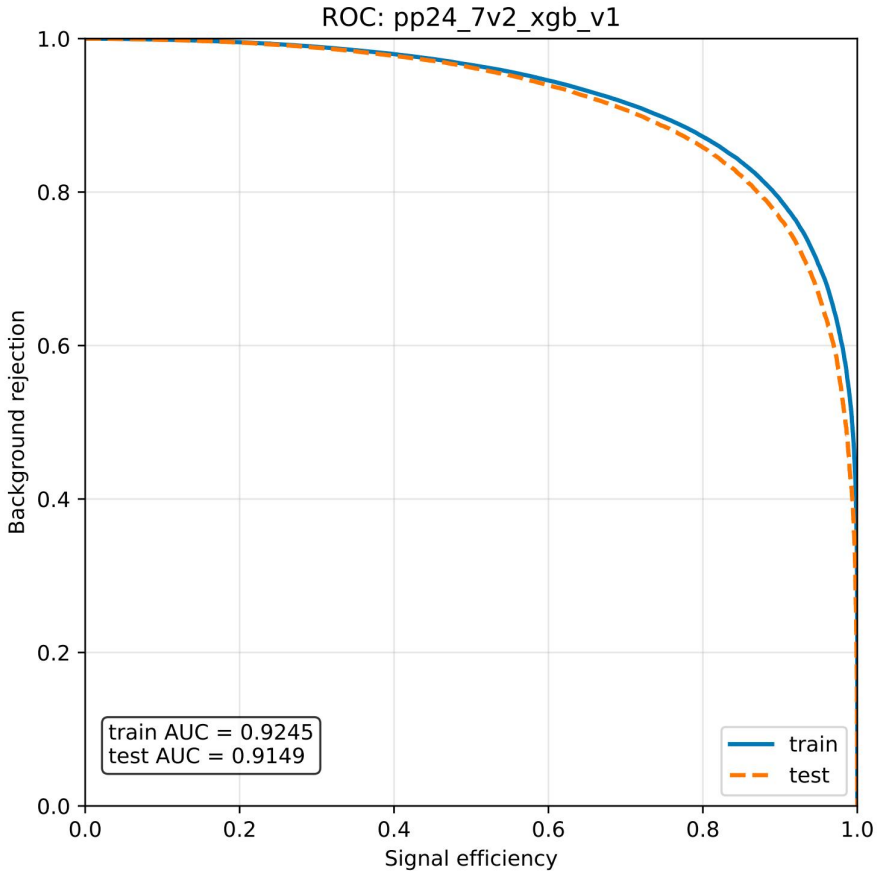


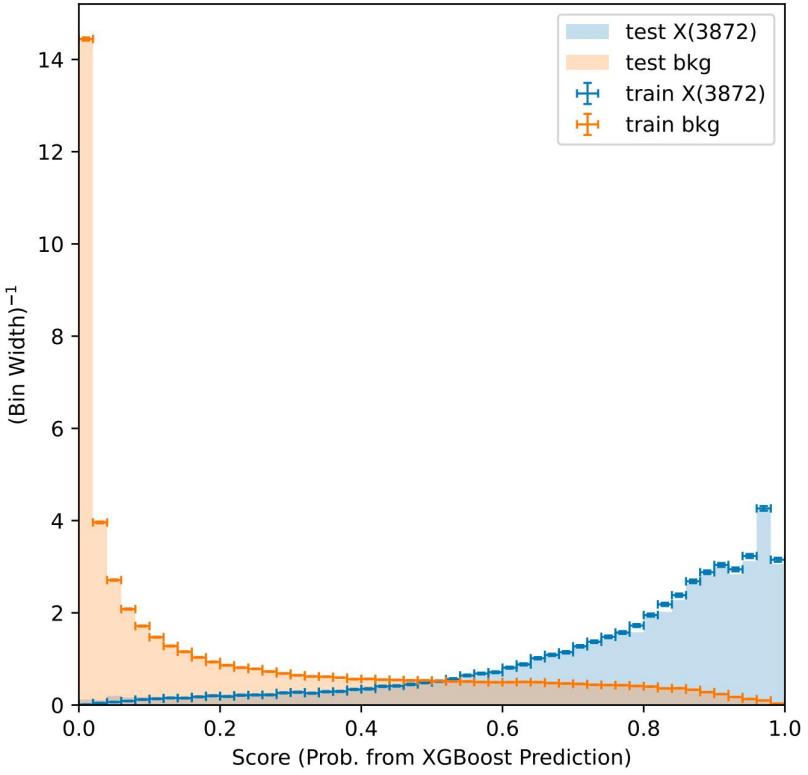




XGBoost Training Results

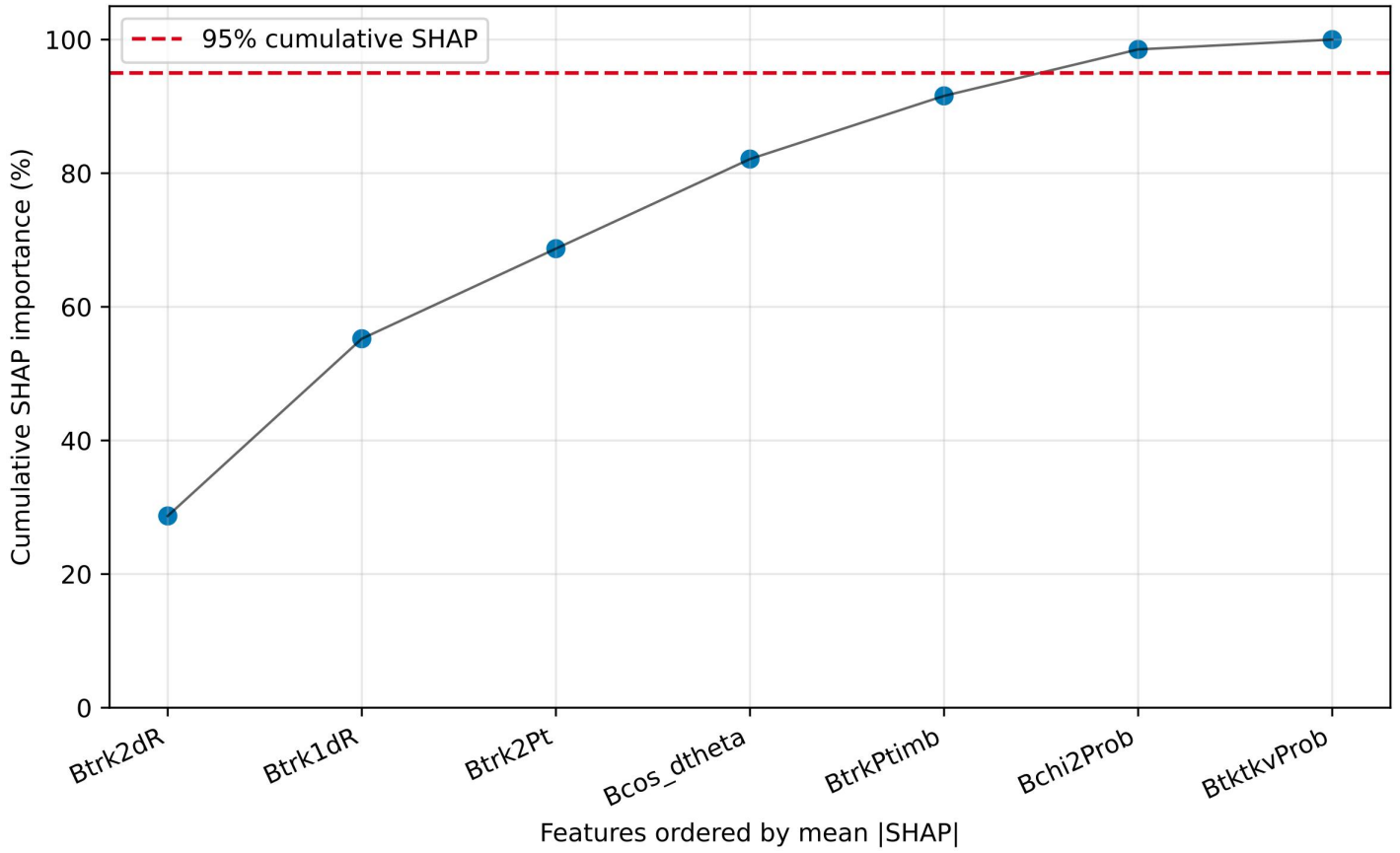
7var: Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob, BtktkvProb





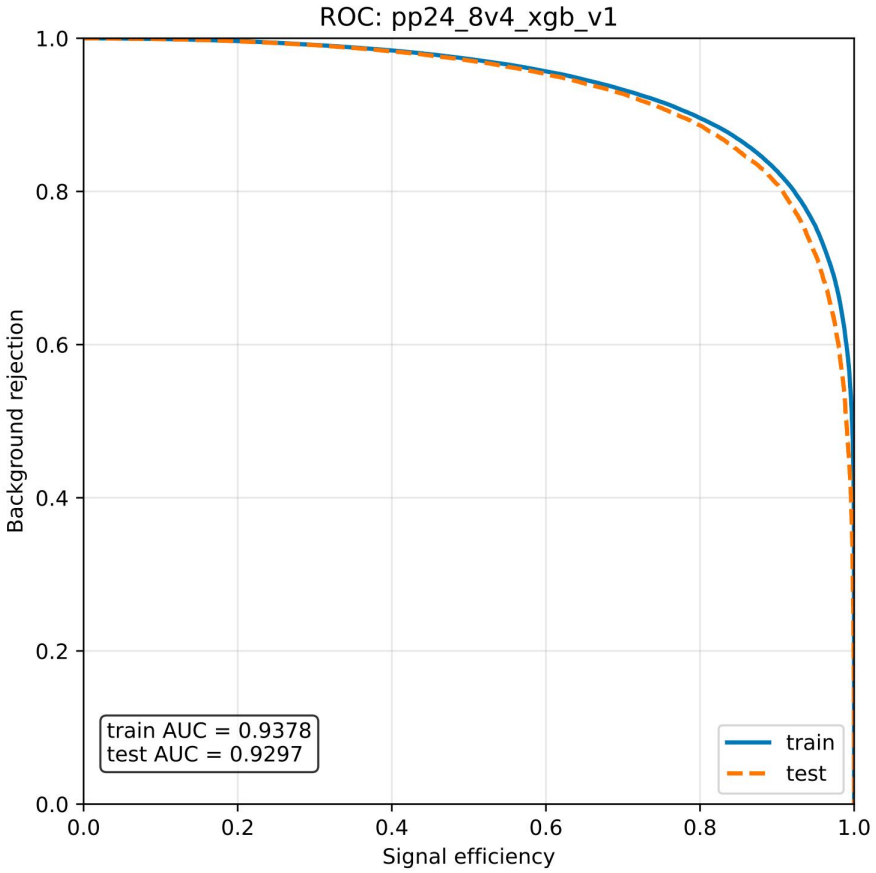
XGBoost Training Results

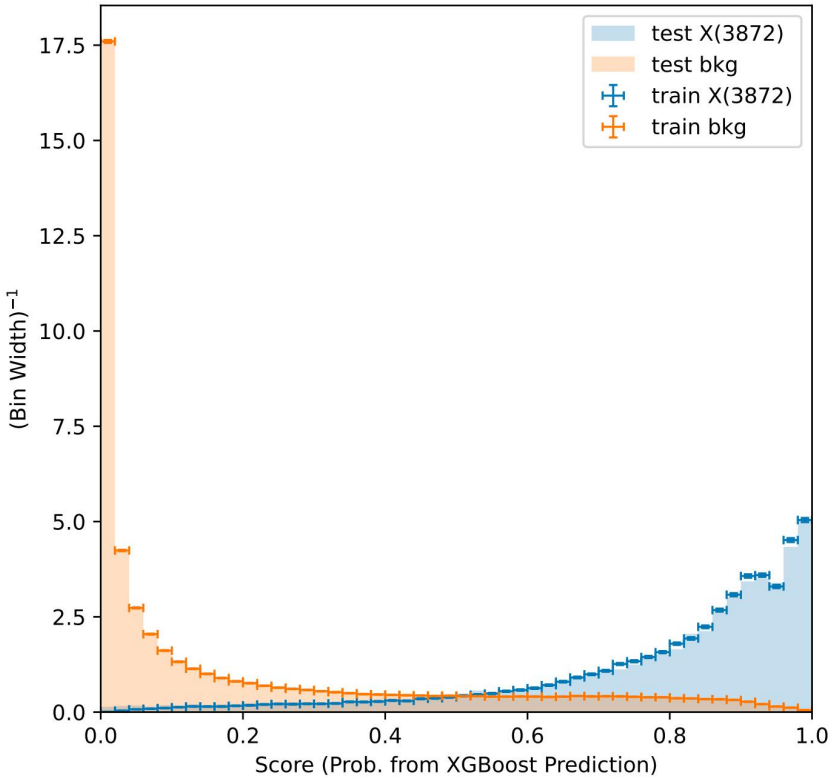
7var: **[**Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob**]**, BtktkvProb



XGBoost Training Results

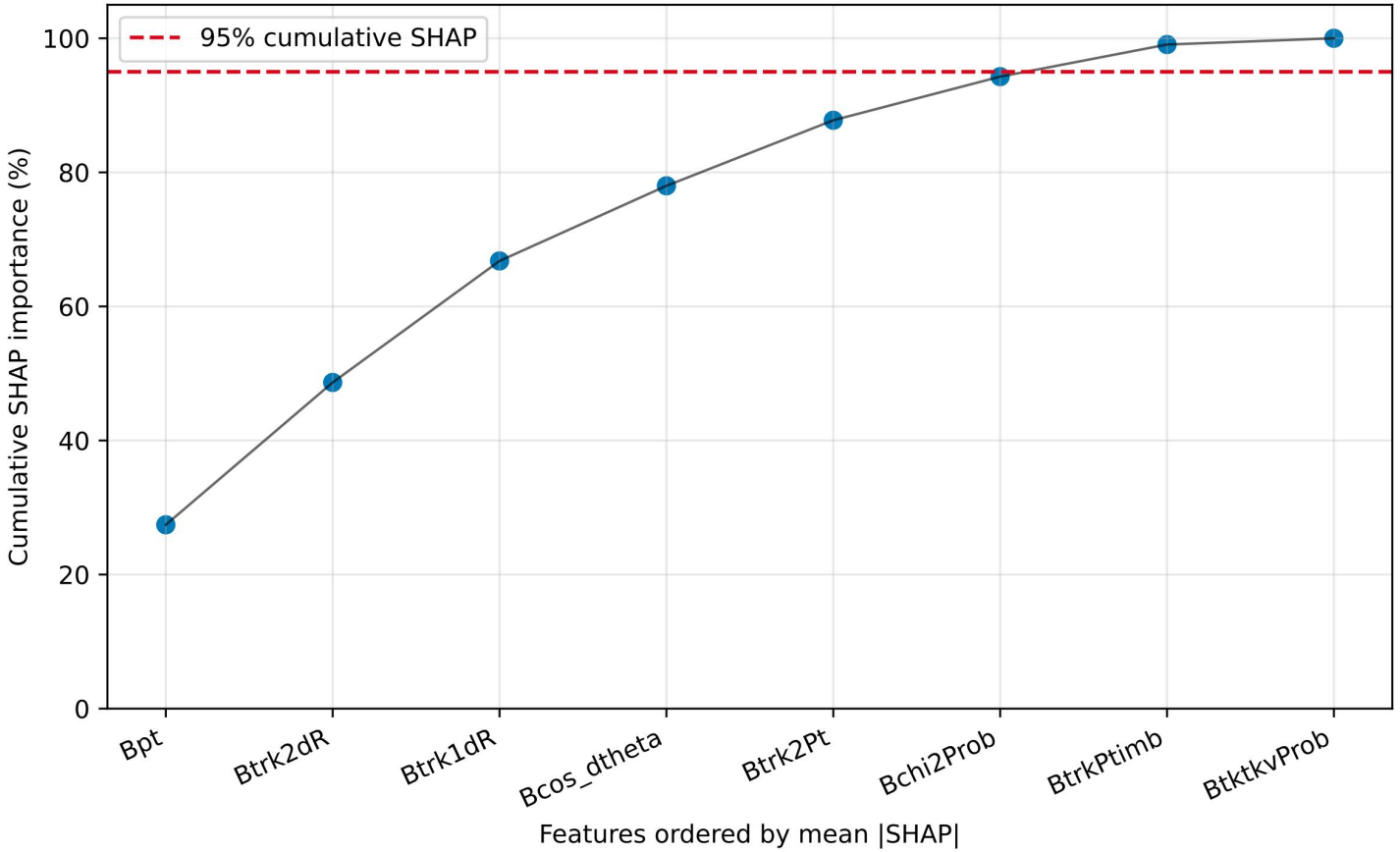
8var: Bpt, Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob, BtktkvProb





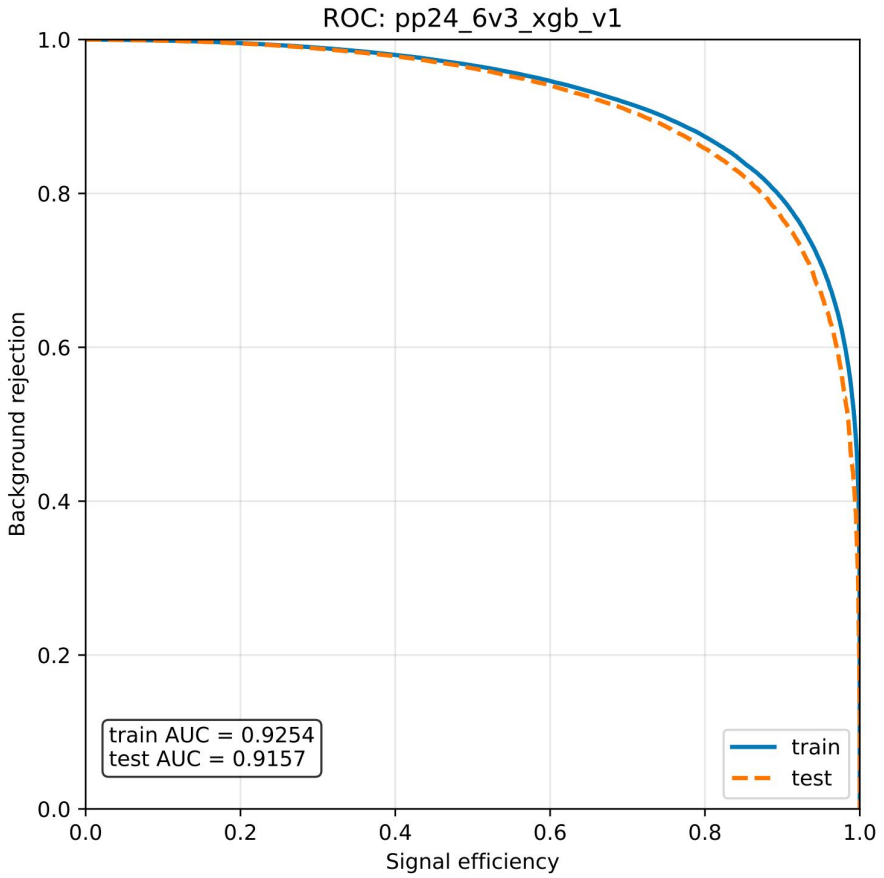
XGBoost Training Results

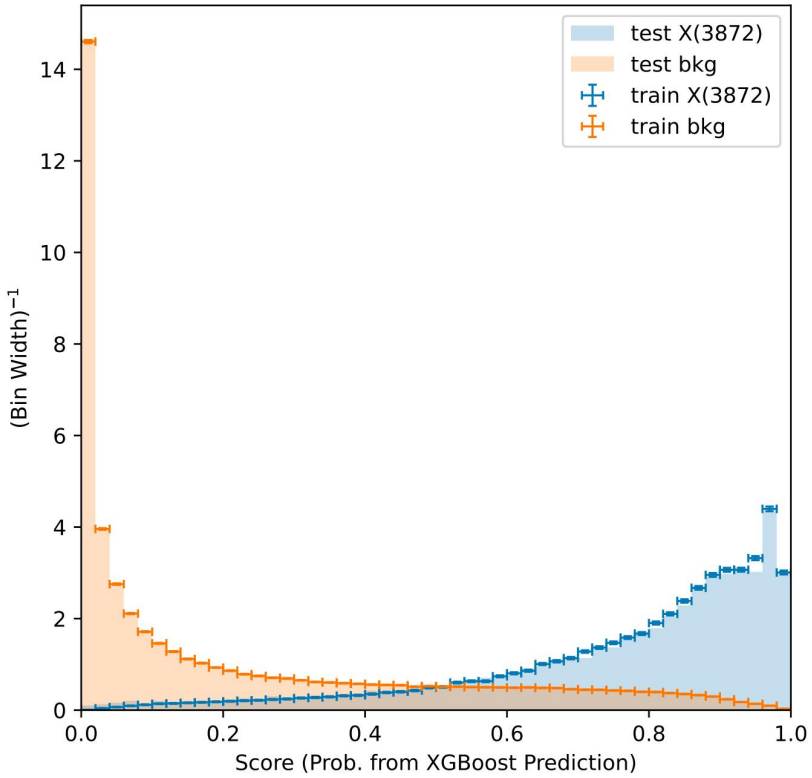
8var: **[**Bpt, Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob**]**, BtktkvProb



XGBoost Training Results

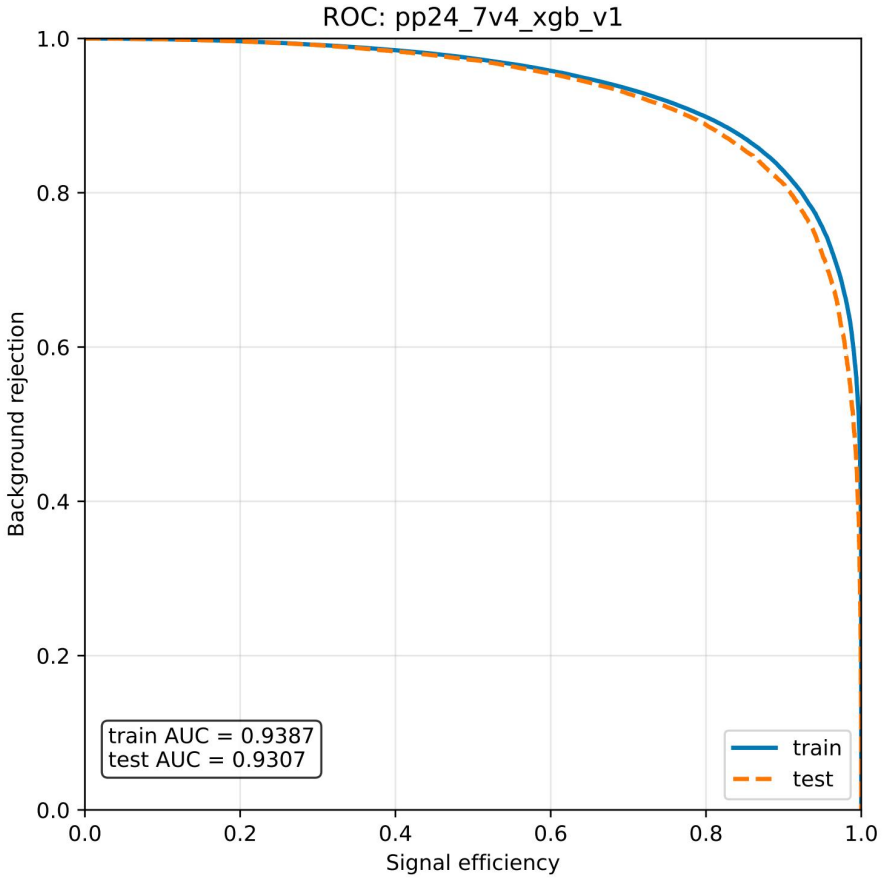
6var (<-7var): Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob

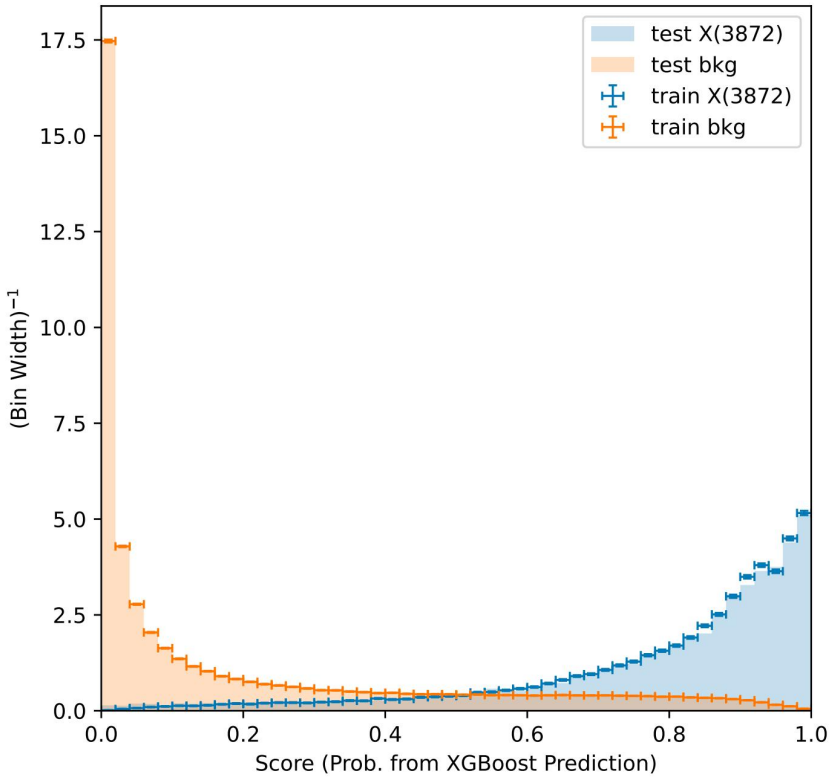




XGBoost Training Results

7var(<-8var): Bpt, Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob





Optimization

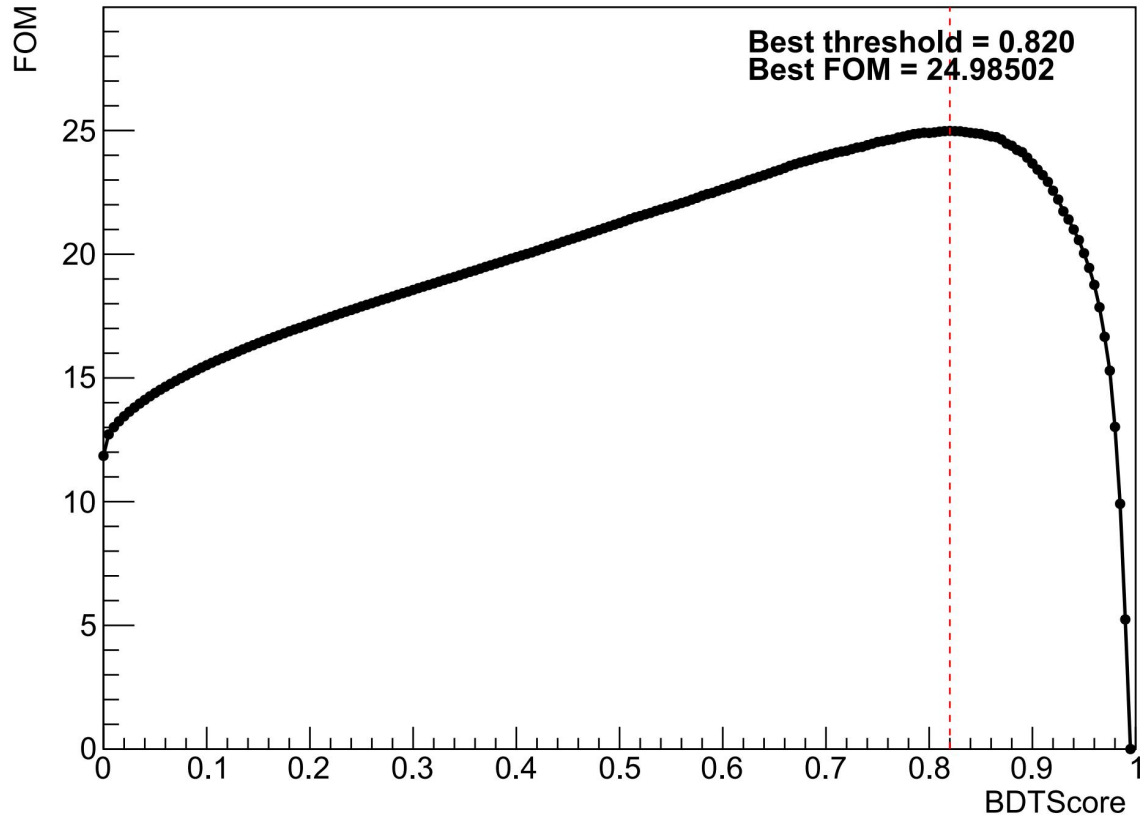
- Goal : discriminate signal from background
- Method : compare the performance and figure of merit (FOM) from different cuts
  - The max FOM of **X(3872)** is considered as **“best cut”**.
  - sideband region: (Bmass > 3.95 && Bmass < 4.0) || (Bmass > 3.75 && Bmass < 3.80)

$$FOM = \frac{f_S \times S_{MC}}{\sqrt{f_S \times S_{MC} + f_B \times B}}$$

$$f_S = \frac{S_{data}}{S_{MC}} \qquad f_B = \frac{B(signal\ region)}{B(sideband\ region)}$$

Optimization

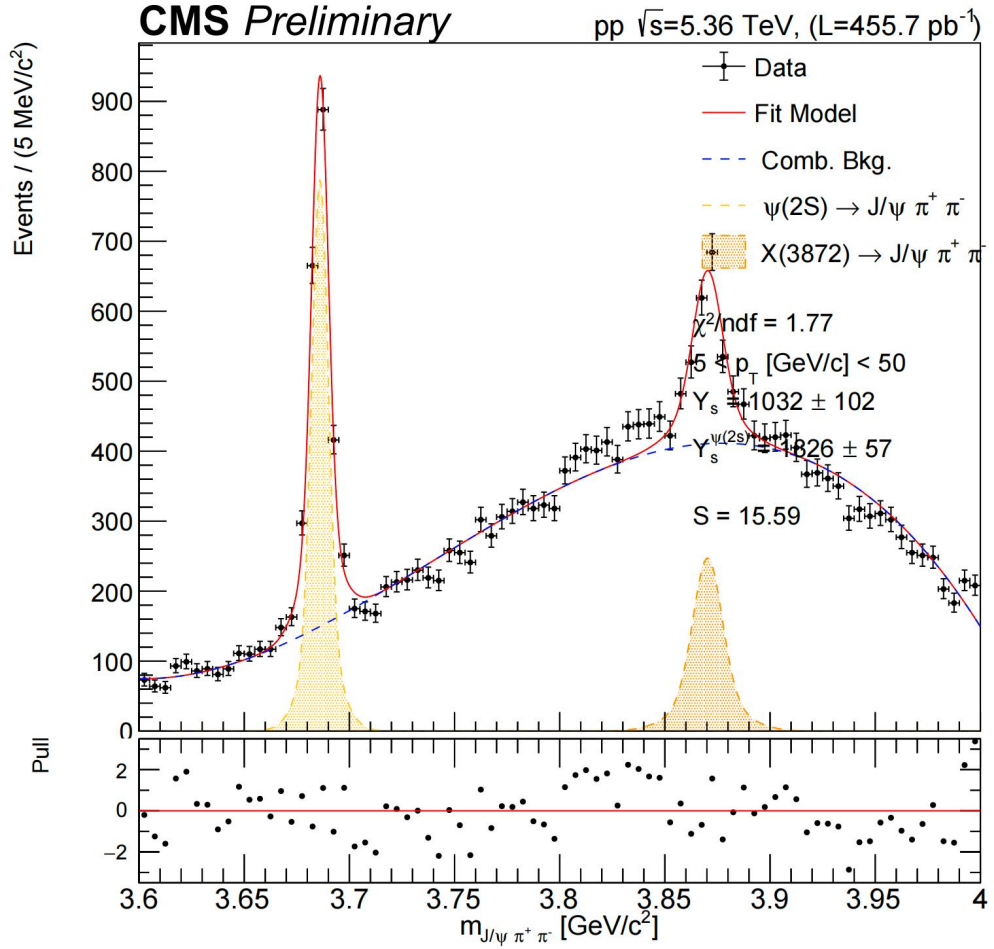
6var (<-7var): Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob



best BDTscore cut = 0.82

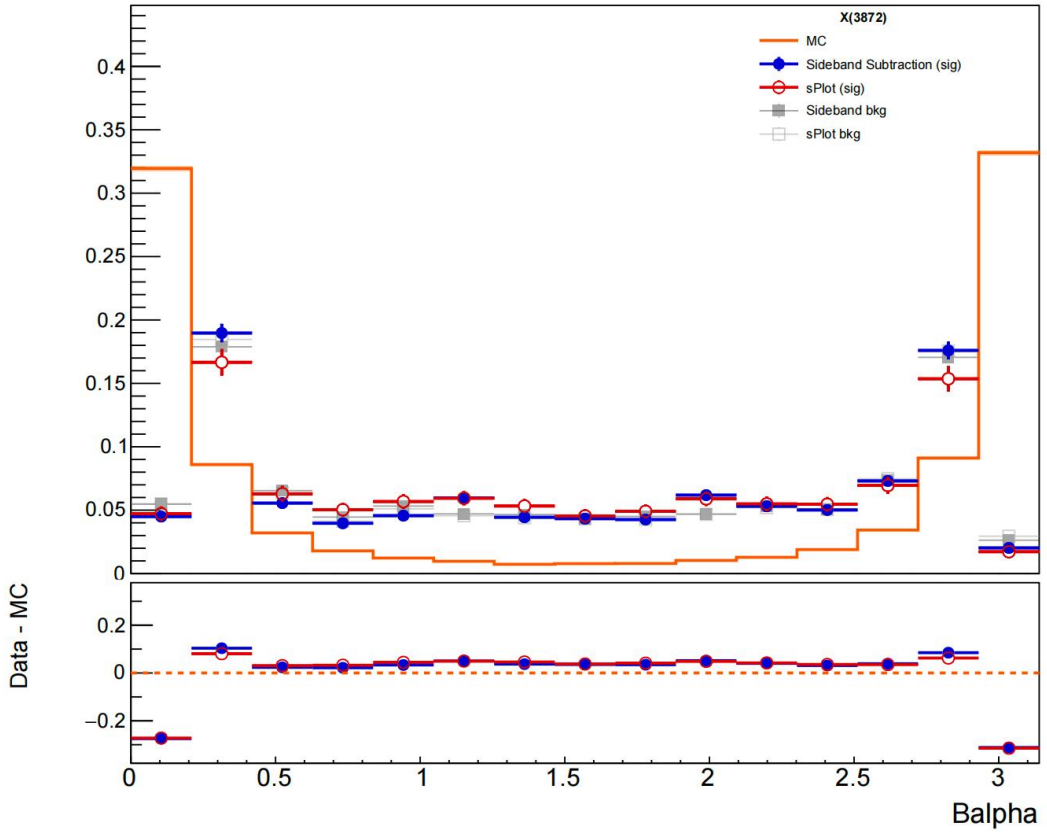
Fit with Optimal Cut

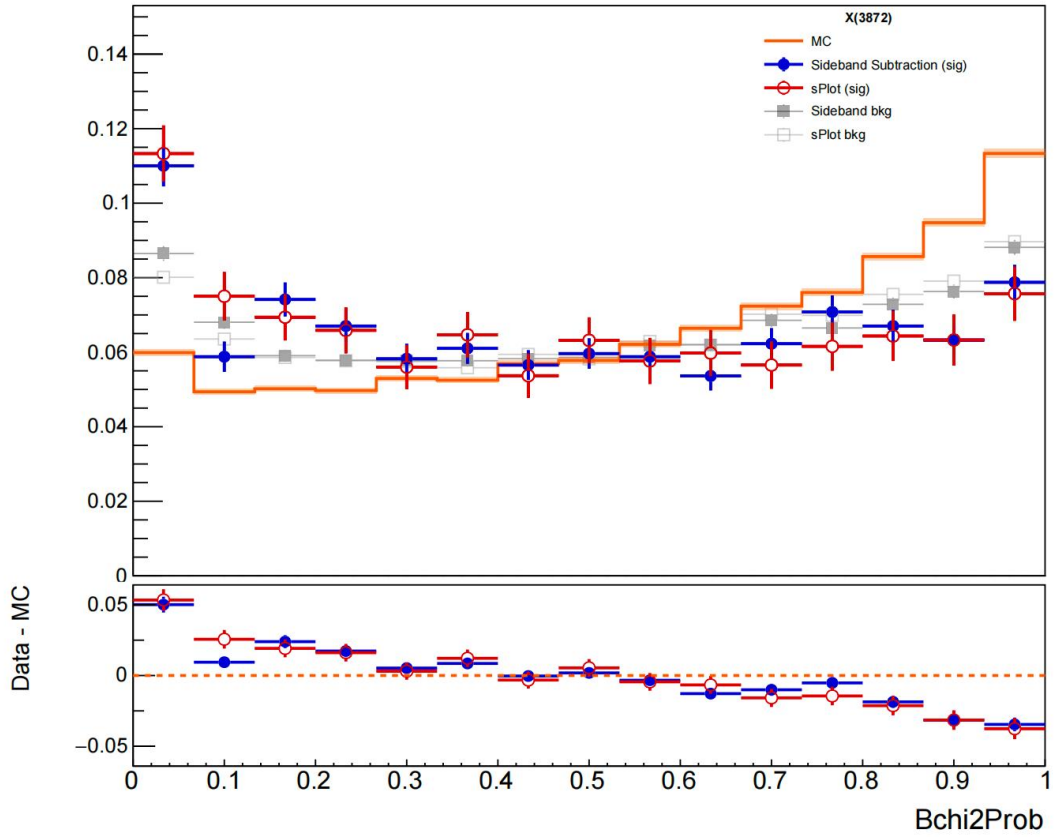
6var (<-7var): Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob



MC Validation

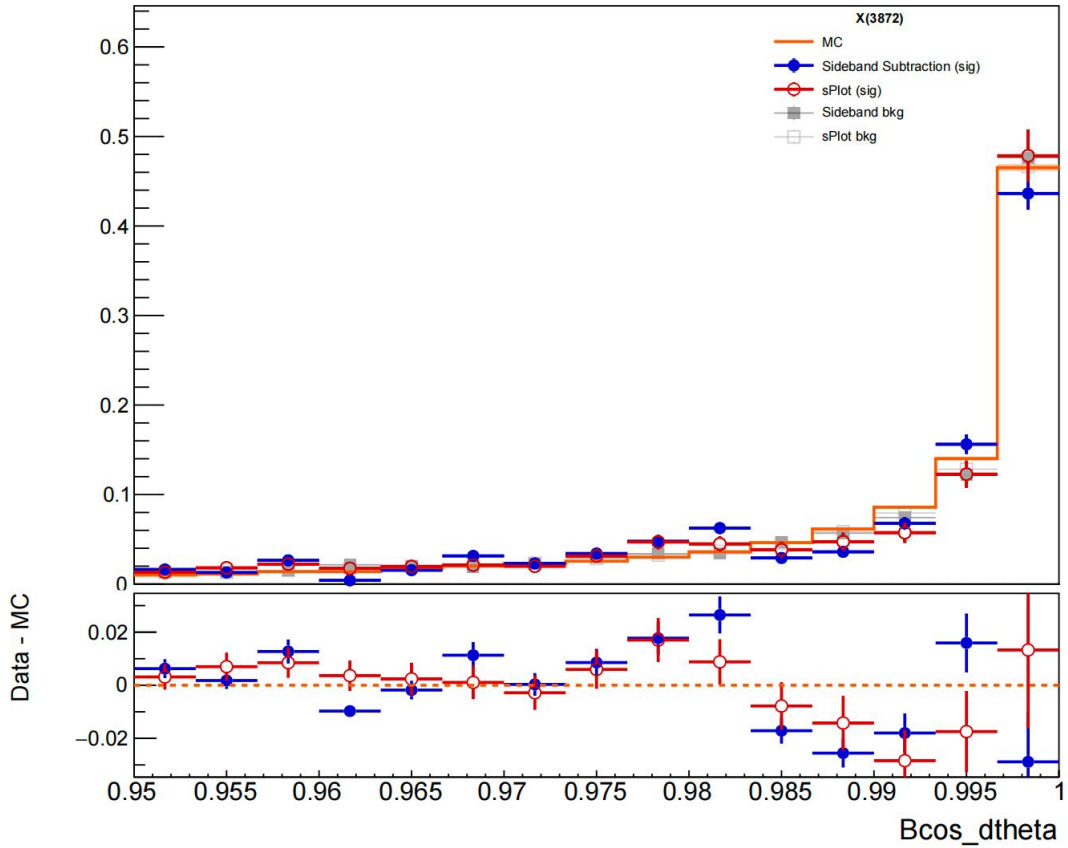
6var (<-7var): Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob

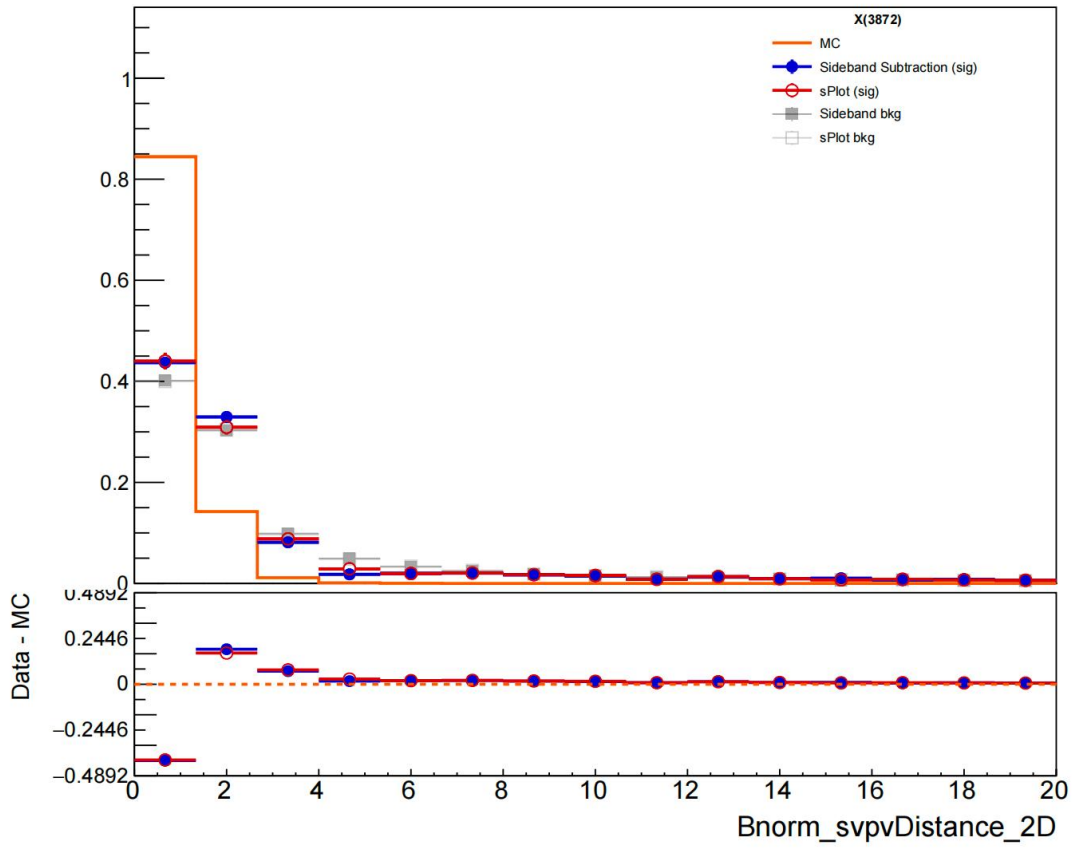




MC Validation

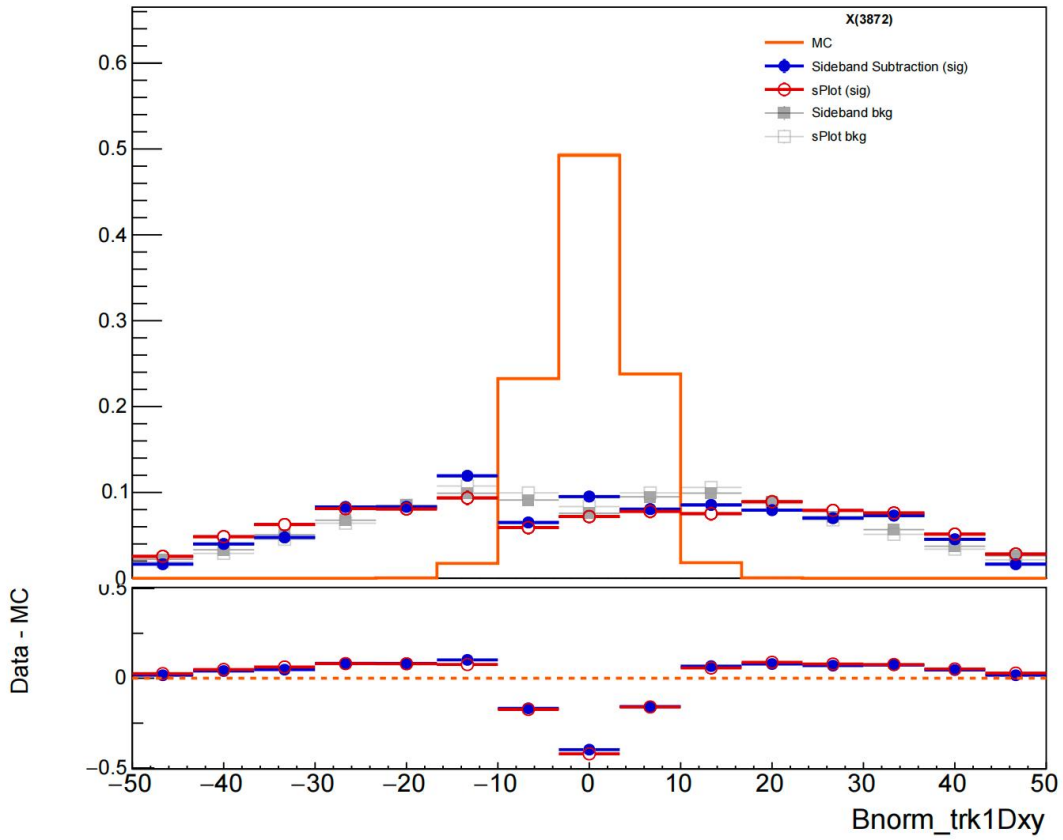
6var (<-7var): Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob

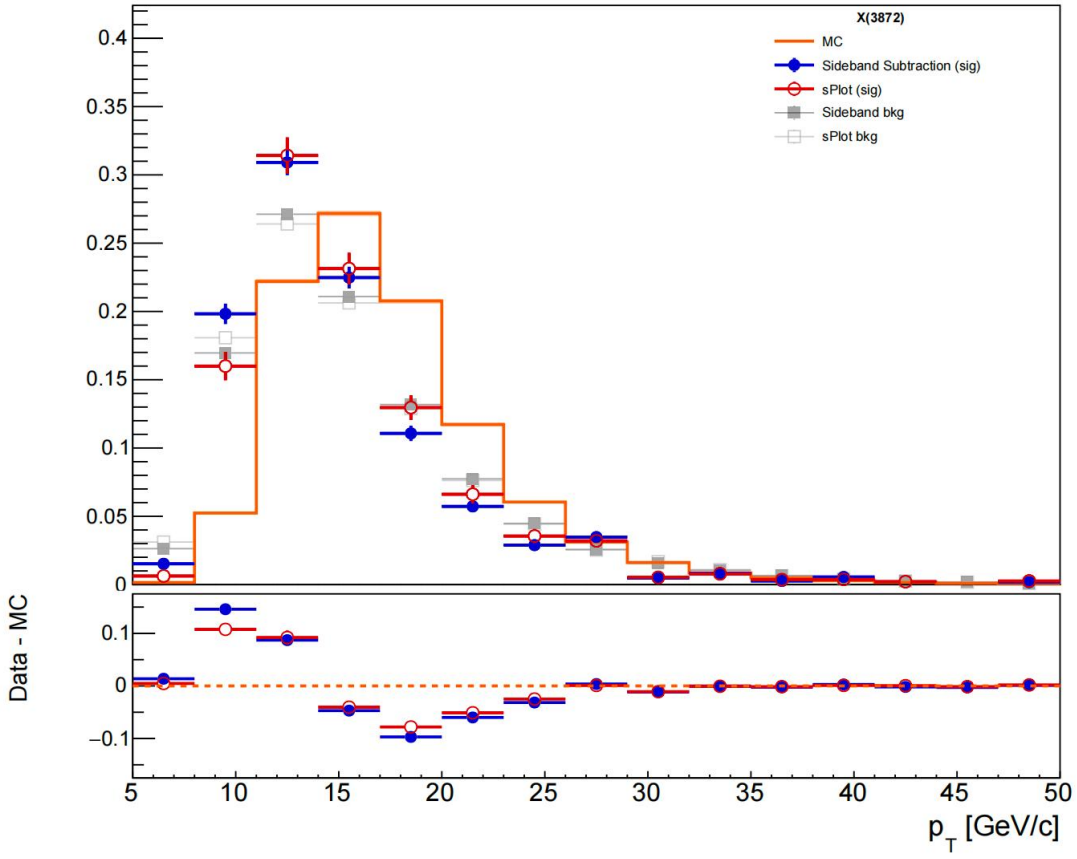




MC Validation

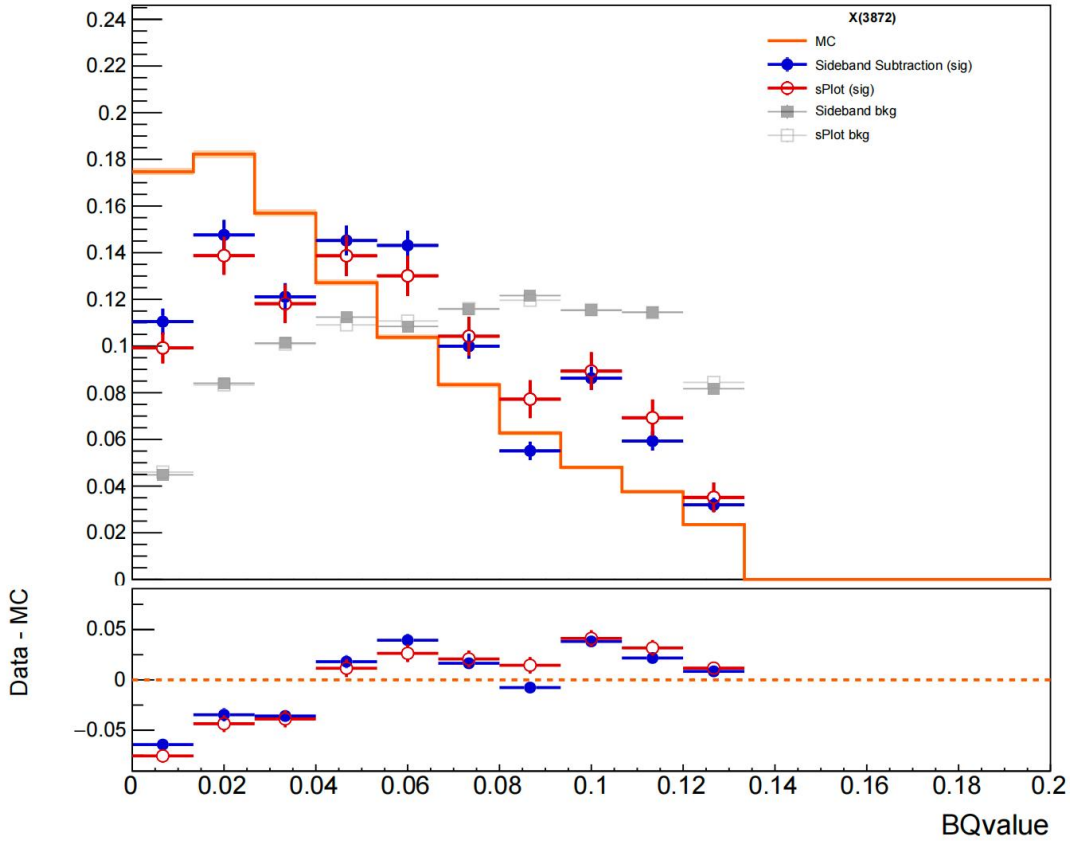
6var (<-7var): Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob

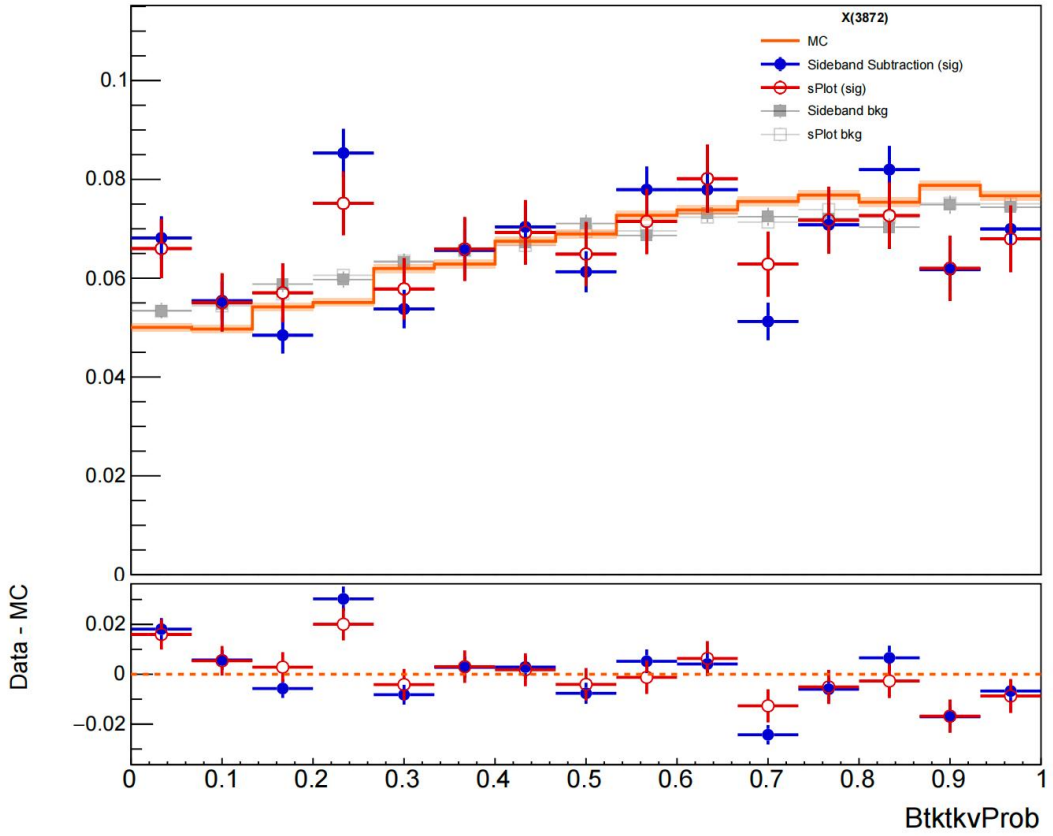




MC Validation

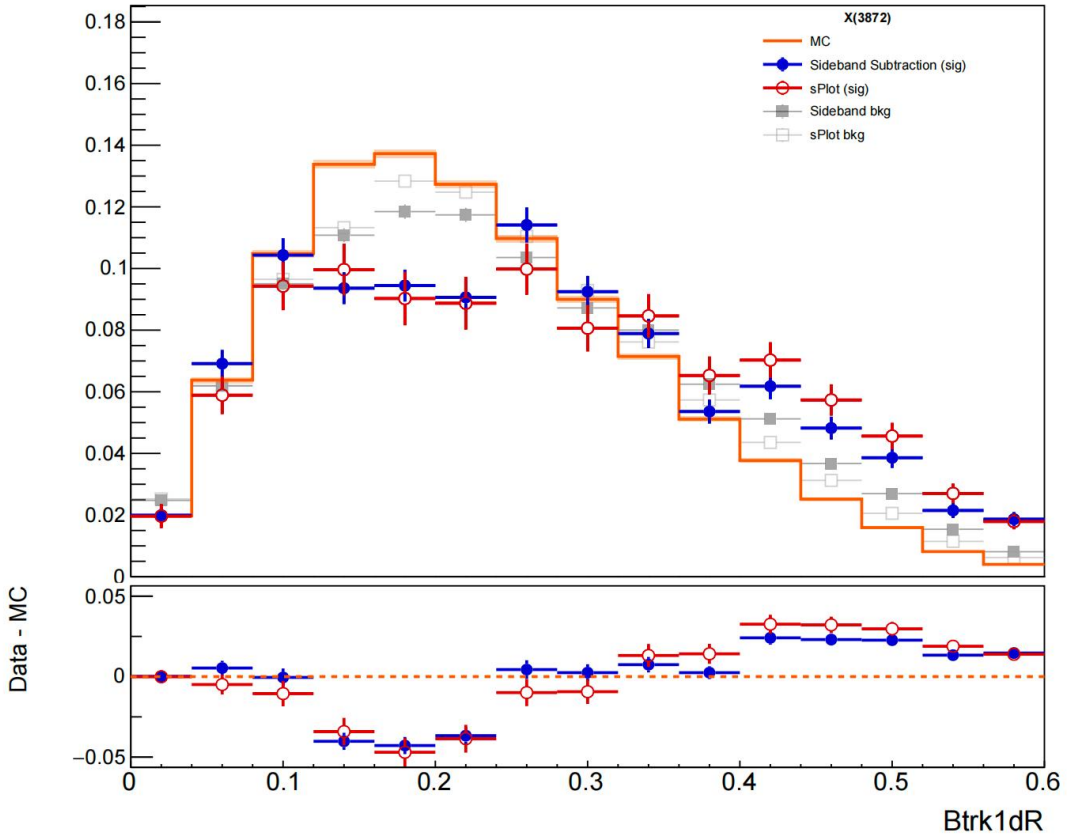
6var (<-7var): Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob

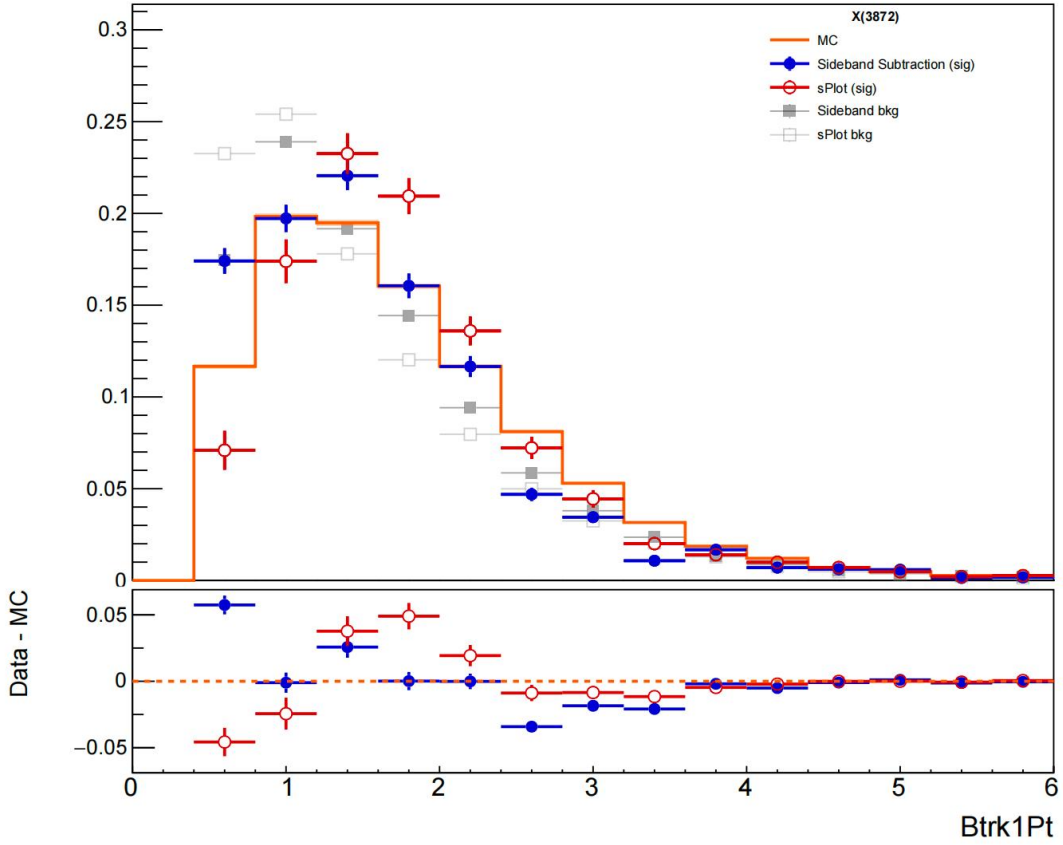




MC Validation

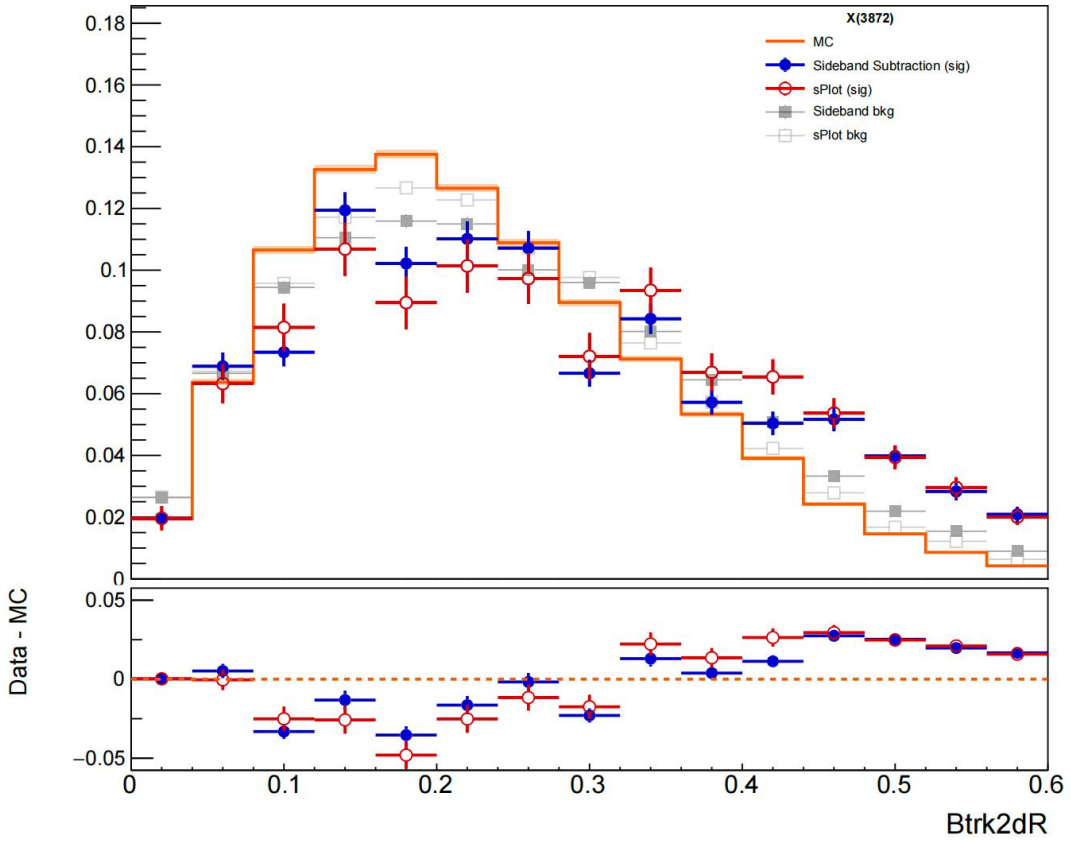
6var (<-7var): Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob

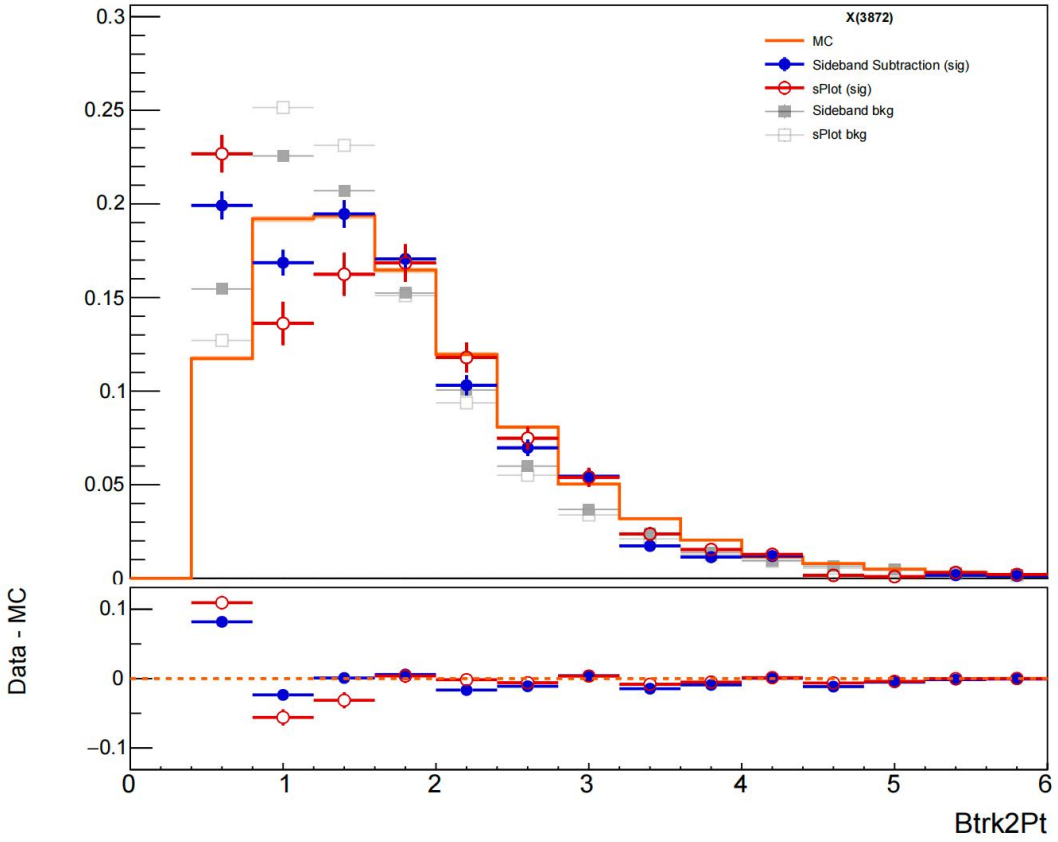




MC Validation

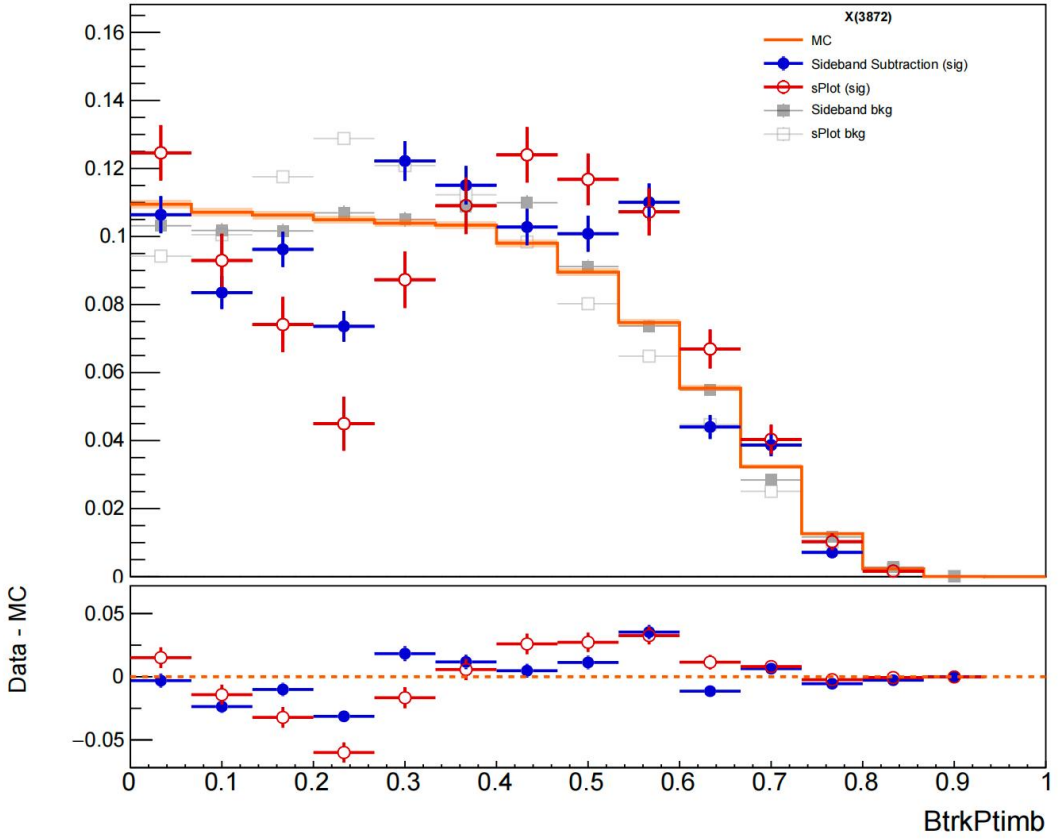
6var (<-7var): Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob





MC Validation

6var (<-7var): Btrk1dR, Btrk2dR, BtrkPtimb, Btrk2Pt, Bcos_dtheta, Bchi2Prob



Thank you